PLATFORM TECHNOLOGIES

OBJECTIVES:

- Define platform technologies and explain their role in modern computing.
- Differentiate between hardware, software, and digital platforms.
- Identify common examples of platform technologies.
- Discuss the importance of platforms in innovation and interoperability.

WHAT DOES PLATFORM TECHNOLOGIES MEAN?

- A foundation of hardware, software, and services upon which applications or processes are built.
- A platform technology serves as the foundation for DEVELOPING and RUNNING BUSINESS APPLICATIONS.
- The platform includes analytics and data management tools and requires a COMBINATION OF SOFTWARE AND HARDWARE COMPONENTS.
- The “bridge” that connects users, developers, and services.

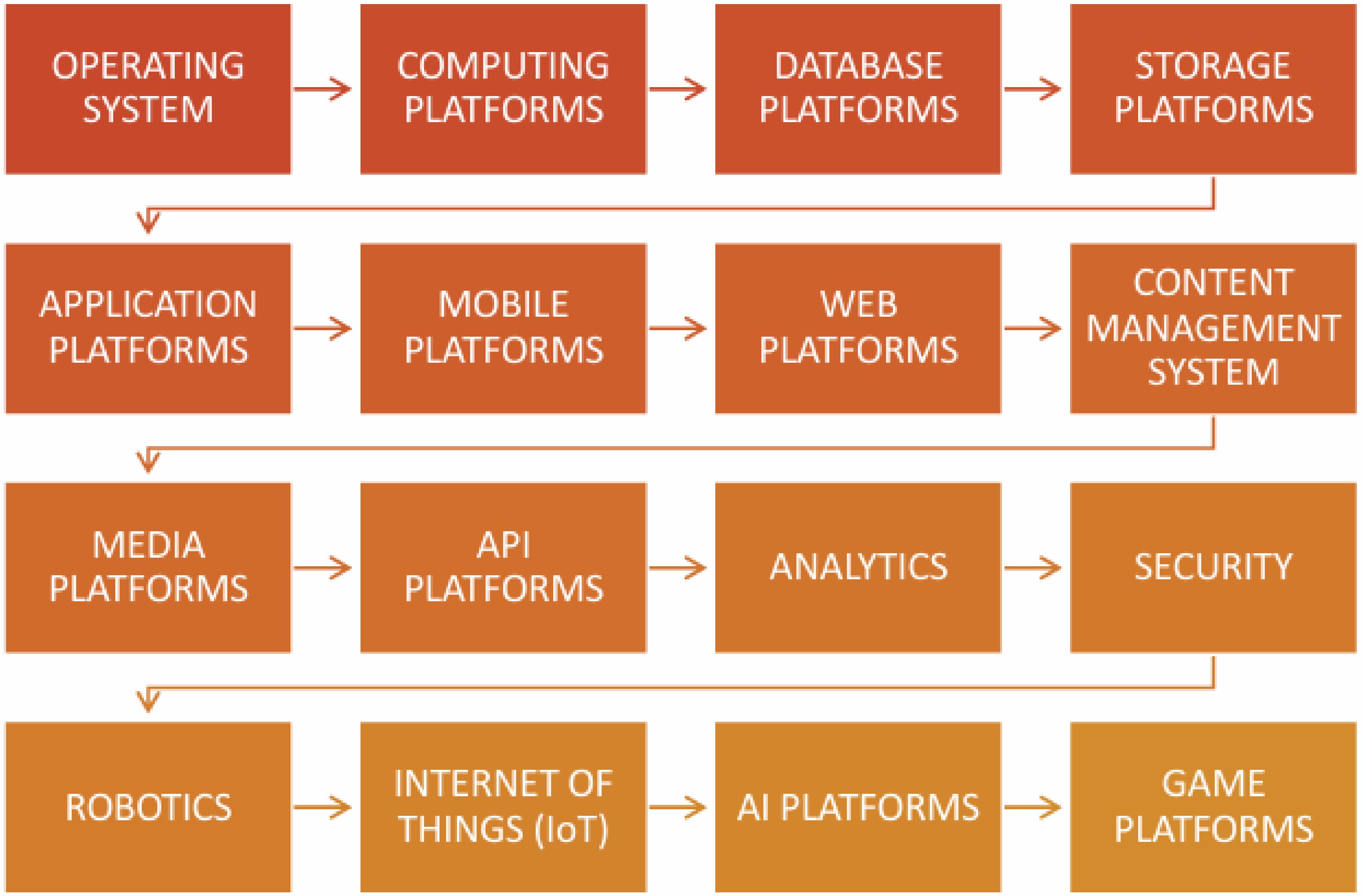
TYPES OF PLATFORM TECHNOLOGIES:

- **Hardware Platforms** – Physical computing infrastructure (IoT devices, Servers)
- **Software Platforms** – Development and execution environment (Windows, JVM, .NET)
- **Digital Service Platforms** – Combines software + hardware for services (AWS, Netflix, OpenAI)
- **Development Platforms** – Tools for app or system creation (Flutter, Unity, TensorFlow)

IMPORTANCE OF PLATFORM TECHNOLOGIES

- Support innovation and scalability
- Enable cross-platform integration
- Facilitate ecosystem growth
- Improve user experience and collaboration
- Without platforms, developers would need to start from scratch every time.

THE FOLLOWING ARE THE COMMON TYPES OF PLATFORM TECHNOLOGIES



OPERATING SYSTEMS

- Operating Systems provide the basic services required to use hardware. These are the lowest level of platform.

COMPUTING PLATFORMS

- Platforms built on top of operating systems that provide computing functionality in areas such as cloud computing and visualization.

DATABASE PLATFORMS

- Cloud platforms for deploying and managing various types of database such as relational, NoSQL and in-memory databases.

STORAGE PLATFORMS

- Platform for scalable storage of objects and files including API's and value-added services such as resilient storage that is backed up in multiple locations.

APPLICATION PLATFORMS

- Application platform are environments and toolkits for developing and deploying applications, a class of software that is primarily designed to be used by people.

MOBILE PLATFORMS

- Mobile Platforms include mobile operating systems and environments for building mobile apps. They also include cloud platforms for building mobile backends that provide services to mobile apps. This may include specialized API's that are useful for mobile app developers in areas such as location services and voice recognition

WEB PLATFORMS

- Platforms that provide services that are useful to websites and web-based software as a service such as web servers, web application servers, content delivery networks and edge computing

MEDIA PLATFORMS

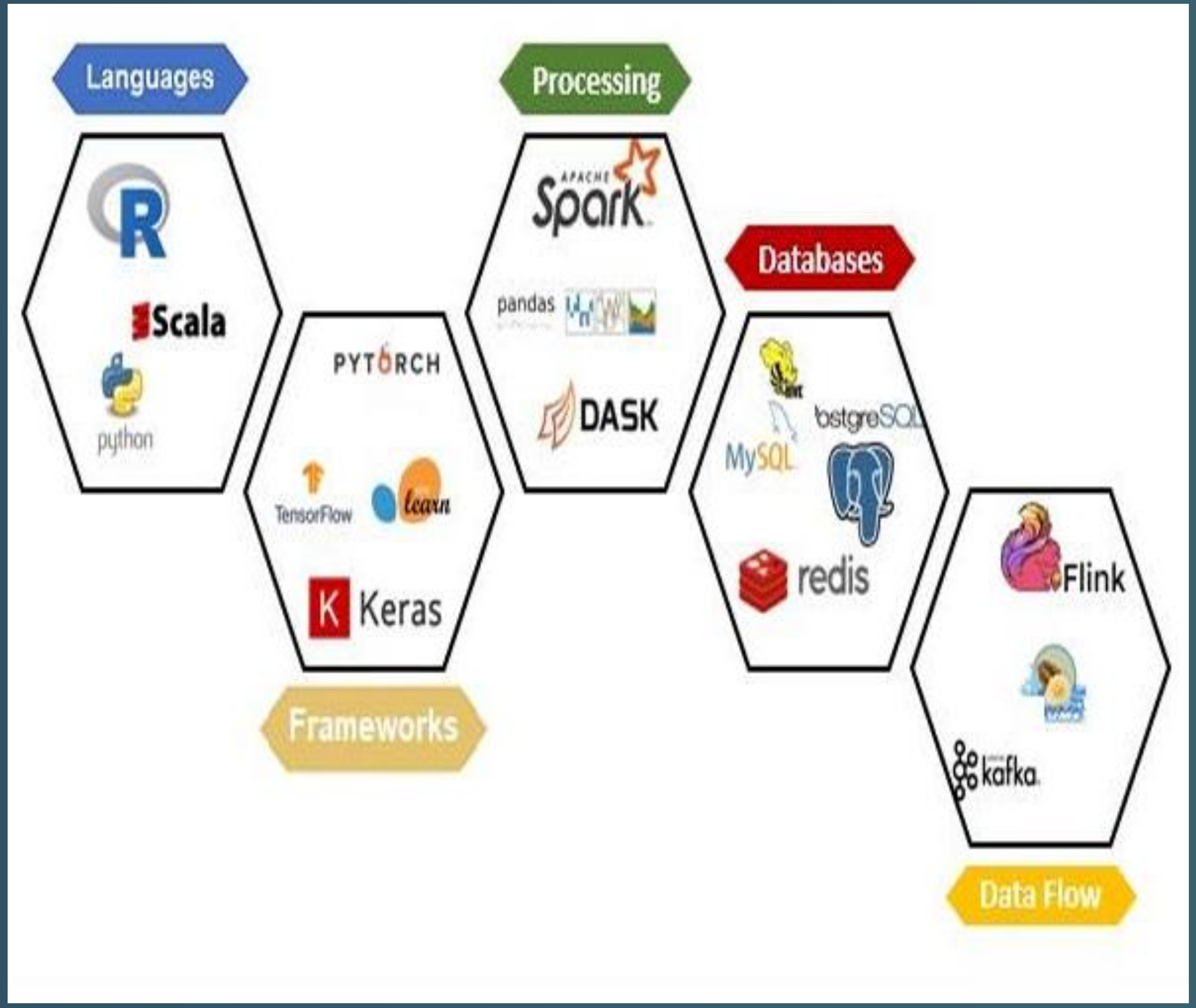
- Platform for media publishing and analysis with tools such as video transcoding streaming and recognition.

API PLATFORMS

- Cloud platforms for deploying Application Programming Interface (API) that are typically built around an API getaway that performs functions such as load balancing, latency and reduction and rate limiting.

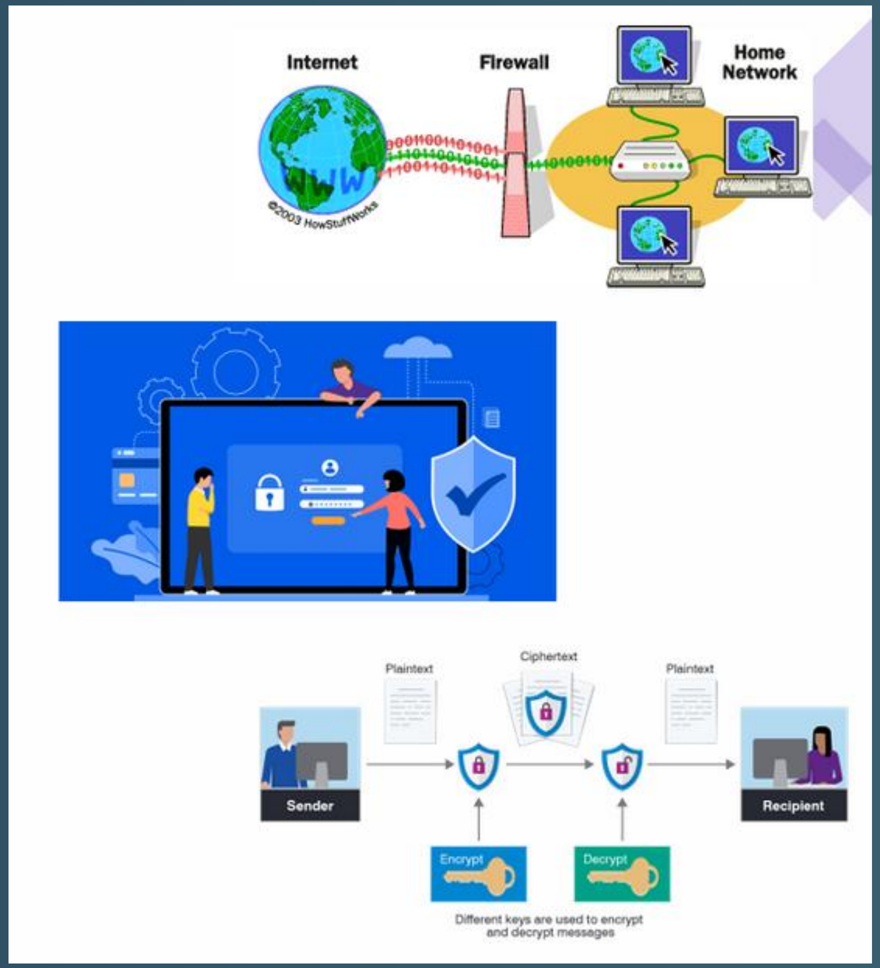
ANALYTICS

- Services for capturing, processing, analyzing and visualizing data. This may include tools for ingesting, processing, querying and managing big data.



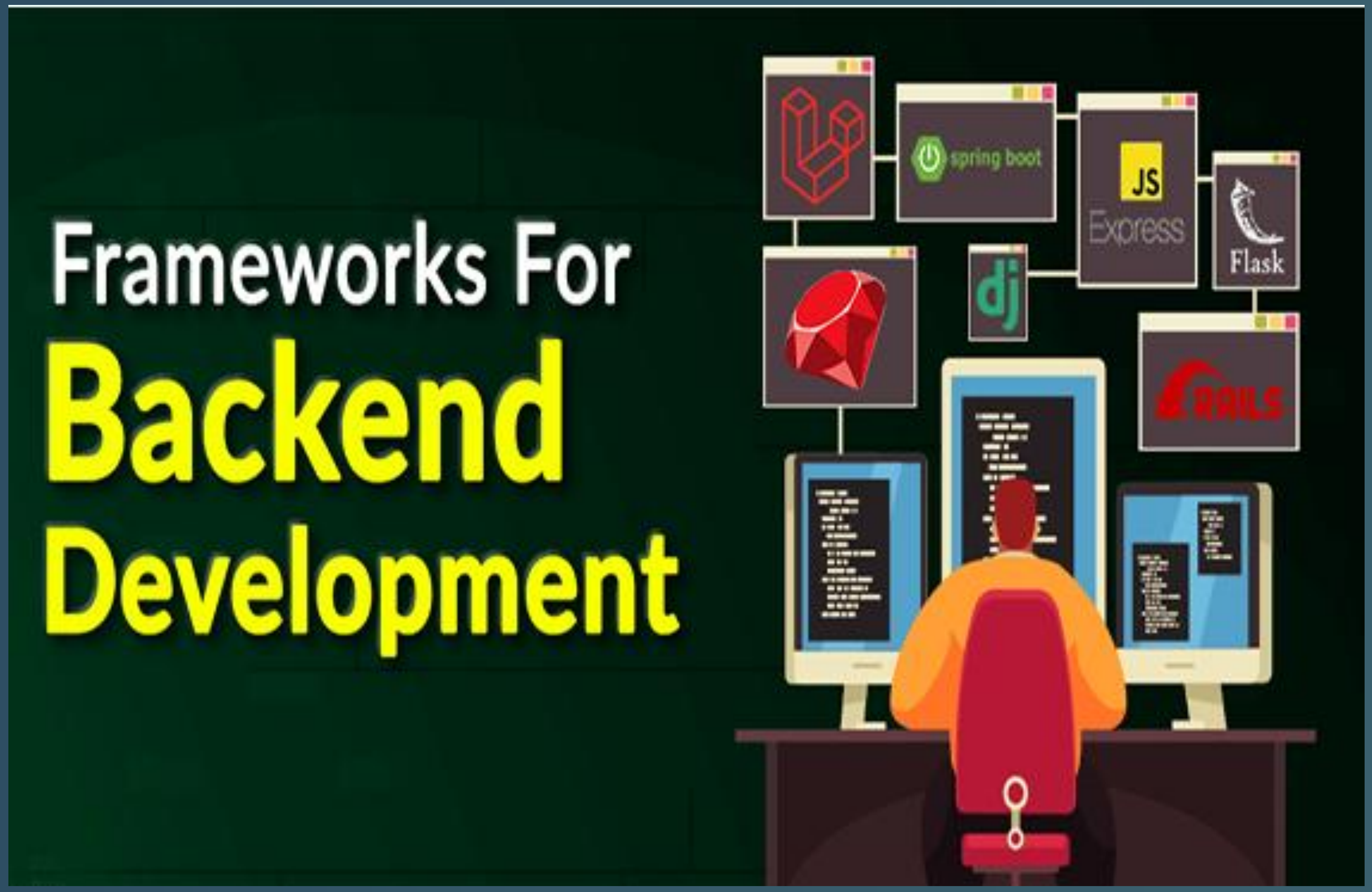
SECURITY

- Security Services such as firewalls, identity & access management, directory services, certificates, compliance reporting, encryption, key management and threat detection



ROBOTICS

- Robotics platforms may include an operating system for robots with framework for developing and deploying backend systems and services for robots on cloud infrastructure.



INTERNET OF THINGS (IOT)

- Internet of Things platforms may include an operating system for devices and a cloud platform with specialized API's for internet of things areas such as device management, IoT security and analytics

ARTIFICIAL INTELLEGENCE (AI) PLATFORMS

- Services that are based on AI such as voice synthesis service and tools for building your own AI such as a Machine Learning API. This may also include environments for running your AI that are optimized for machine learning such as machine learning databases.

GAME PLATFORMS

- Environments that are optimized for running game services such as backends for mobile games or massively multiplayer online games. These may include services such as 3D game engines, AR and VR API's.

TAKEAWAYS:

• Platform technologies are the backbone of digital innovation.
• They connect users, developers, and systems through shared infrastructures.
• Platforms promote efficiency, scalability, and collaboration.
• The future of technology depends on interconnected platforms in AI, cloud, and IoT.

ACTIVITIES

1.

- List 3 platform technologies you use daily.

2.

- Classify each as: Hardware Platform, Software Platform, or Digital Service Platform.

3.

How do these platforms make your daily tasks easier or more efficient?

"EVERY TECHNOLOGY WE USE STANDS ON A PLATFORM THAT CONNECTS PEOPLE, DEVICES, AND IDEAS."